David Larios, Ph.D.

QUANTITATIVE BIOPHYSICIST | PROTEIN ENGINEERING | HIGH-THROUGHPUT PLATFORM DEVELOPMENT

Pasadena, CA | dlarios22@gmail.com | linkedin.com/in/david-larios-phd | Project website

PROFILE

Quantitative biophysicist who builds high-throughput platforms to dissect and engineer protein function from sequence and structure. Built an automated cell-free platform capable of screening up to 1,152 proteins per run, revealing previously unknown functional diversity in an undercharacterized protein family and enabling the rational engineering of a variant with an enhanced functional profile.

RESEARCH EXPERIENCE

Graduate Researcher | Phillips & Thomson Labs, Caltech | Pasadena, CA 2019 - 2026

- Built and validated an automated high-throughput platform to screen up to 1,152 kinesin protein variants directly from DNA in cell-free droplets, eliminating individual protein purification and integrating confocal microscopy with quantitative image analysis.
- Applied the platform to reveal previously unrecognized functional diversity within the kinesin family, identifying three quantitatively distinct classes of protein activity.
- Expanded the screening workflow by integrating biochemical assays with molecular dynamics simulations of AlphaFold-predicted structures, identifying a molecular basis for the functional diversity revealed by the platform.
- Designed and screened a complete combinatorial library of chimeric proteins to dissect the structural basis of protein function, engineering a variant that combined properties from two functionally opposing proteins to achieve enhanced kinetics relative to the original library.

Research Intern | Elowitz Lab, Caltech | Pasadena, CA Jun - Aug 2018

- Engineered fluorescent human iPSC reporter lines using lentiviral delivery and quantified dynamic Wnt signaling through live-cell time-lapse microscopy.
- Defined signaling requirements for mesoderm differentiation using pathway perturbations, immunofluorescence, and quantitative image analysis.

Research Intern | Guttman Lab, Caltech | Pasadena, CA Jun - Aug 2017

- Expressed, purified, and validated a recombinant protein from E. coli using nickel-affinity chromatography, SDS-PAGE, and Western blotting.
- Developed a covalent RNA-protein capture workflow combining CRISPR/Cas9 genome editing with UV crosslinking to isolate interactions from human cells with low background.

TECHNICAL SKILLS

Protein engineering & molecular biology: combinatorial chimera-library design and construction, (Gibson/Golden Gate) assembly, linear DNA template preparation, cell-free (TXTL) expression, Ni-NTA purification, SDS-PAGE/Western, ATPase and enzyme kinetics

High-throughput screening & imaging: cell-free droplet screening, plate-based automation, confocal and time-lapse microscopy, quantitative assay development

Computational analysis: Python, statistical analysis, image analysis (PIV), AlphaFold, molecular dynamics, statistics for large screens (clustering, dimensionality reduction, multiple-testing correction), git (HPC/SLURM)

Cell engineering: CRISPR/Cas9, lentiviral delivery, human iPSC culture and differentiation, immunofluorescence, live-cell signaling assays

EDUCATION

Ph.D., Biophysics | California Institute of Technology (Caltech) June 2026

B.Sc., Genomic Biotechnology | Autonomous University of Nuevo León 2018

SELECTED WORK

Larios, D. A. C. et al. Reverse Engineering the Programming Logic of Cytoskeletal Dynamics. bioRxiv preprint; manuscript under review. Project website | bioRxiv

Qu, Z. et al. Persistent Fluid Flows Defined by Active Matter Boundaries. Communications Physics 4, 198 (2021). doi

Lin, Z. et al. Boundaries Program Deformation in Isolated Active Networks. arXiv (2025). arXiv:2510.01713